Cross-presenting human $\gamma\delta$ T cells induce robust CD8$^+$ $\alpha\beta$ T cell responses

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$\gamma\delta$ T cells are implicated in host defense against microbes and tumors but their mode of function remains largely unresolved. Here, we have investigated the ability of activated human Vγ9Vδ2$^+$ T cells (termed $\gamma\delta$ T-APCs) to cross-present microbial and tumor antigens to CD8$^+$ $\alpha\beta$ T cells. Although this process is thought to be mediated best by DCs, adoptive transfer of *ex vivo* antigen-loaded, human DCs during immunotherapy of cancer patients has shown limited success. We report that $\gamma\delta$ T-APCs take up and process soluble proteins and induce proliferation, target cell killing and cytokine production responses in antigen-experienced and naïve CD8$^+$ $\alpha\beta$ T cells. Induction of APC functions in Vγ9Vδ2$^+$ T cells was accompanied by the up-regulation of costimulatory and MHC class I molecules. In contrast, the functional predominance of the immunoproteasome was a characteristic of $\gamma\delta$ T cells irrespective of their state of activation. $\gamma\delta$ T-APCs were more efficient in antigen cross-presentation than monocyte-derived DCs, which is in contrast to the strong induction of CD4$^+$ $\alpha\beta$ T cell responses by both types of APCs. Our study reveals unexpected properties of human $\gamma\delta$ T-APCs in the induction of CD8$^+$ $\alpha\beta$ T effector cells, and justifies their further exploration in immunotherapy research.

anti-microbial immunity | antigen cross-presentation

Immunity to many pathogens and tumors involves major histocompatibility complex class I (MHC I) restricted, cytotoxic CD8$^+$ $\alpha\beta$ T cells, which kill affected leukocytes and nonhematopoietic tissue cells. Microbes and tumors frequently interfere with antigen processing or presentation and thus inhibit appropriate antigen-presenting cell (APC) function; also, many microbes do not infect APCs. However, dendritic cells (DCs), the prototype professional APCs (1), can take up exogenous material derived from infected cells and tumors and direct these to intracellular compartments with access to the MHC I pathway, a process known as antigen "cross-presentation" (2, 3). Such DCs can trigger expansion and differentiation of microbe/tumor-specific CD8$^+$ $\alpha\beta$ T cells. Natural DC subsets in humans that are specialized in antigen cross-presentation are not well defined.

$\gamma\delta$ T cells are essential constituents of innate anti-microbial and anti-tumor defense, yet their role in adaptive immunity is less clear (4–6). $\gamma\delta$ T cells are a distinct subset of CD3$^+$ T cells featuring T cell receptors (TCRs) that are encoded by Vγ- and Vδ-gene segments (4, 5). In peripheral blood of healthy individuals $\gamma\delta$ T cells make up 2–10% of total T cells, and of these the majority (typically >80%) express Vγ9Vδ2-TCRs. A distinguishing feature, their TCRs are selective for conserved non-peptide compounds of microbial or tumor cell origin, including the isoprenoid metabolites isopentenyl pyrophosphate (IPP) and (*E*)-4-hydroxy-3-methyl-but-2-enyl pyrophosphate (HMB-PP), which are recognized in a MHC-independent fashion (7, 8). In agreement, Vγ9Vδ2$^+$ T cells are highly expanded in patients suffering from microbial infections.

We have recently reported that IPP-stimulation of human blood Vγ9Vδ2$^+$ T cells leads to the expression of lymph node migration receptors and the transformation of these cells into professional APCs, termed $\gamma\delta$ T-APCs, capable of inducing CD4$^+$ T cell responses (9, 10). Of note, antigen-presenting $\gamma\delta$ T cells have also been reported in cows (11); pigs (12); and, most recently, mice (13). Reactivity to HMB-PP-expressing microbes and certain tumors suggested to us a role for human $\gamma\delta$ T-APCs in the induction of pathogen/tumor-specific CD8$^+$ T effector cells. Rapid and uniform activation in response to a single stimulus of IPP or HMB-PP represents a highly useful tool for investigating $\gamma\delta$ T cell functions and allowed us to examine the ability of $\gamma\delta$ T-APCs to cross-present soluble microbial and tumor antigens to CD8$^+$ responder cells.

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Results

Human $\gamma\delta$ T-APCs Efficiently Cross-Present Soluble Proteins to CD8$^+$ $\alpha\beta$ T Cells. First, we examined the ability of $\gamma\delta$ T-APCs to induce $\alpha\beta$ T cell proliferation in response to the complex protein mixture *Mycobacterium tuberculosis* purified protein derivative (PPD). $\gamma\delta$ T-APCs or monocyte-derived DCs were loaded with PPD, washed and then cocultured with autologous, 5- (and 6-) carboxyfluorescein diacetate succinimidyl ester (CFSE)-labeled responder cells. Using bulk CD3$^+$ T cells as responder cells, both CD8$^+$ T cells and CD4$^+$ T cells showed clear proliferation responses, as assessed by reduction in CFSE signals (Fig. 1*A*). Similar antigen-dependent responses were obtained with purified naïve CD8$^+$ $\alpha\beta$ T cells as responder cells.

To confirm these initial findings in support of cross-presentation by $\gamma\delta$ T-APCs, we turned to an experimental model that allowed more detailed investigations. This model included the well defined influenza virus-encoded matrix protein M1 that induces strong CD8$^+$ $\alpha\beta$ T cell responses to M1p58–66, the immunodominant peptide contained within M1, in HLA A*0201 (HLA-A2)-positive individuals (14). First, cross-presentation was studied in a HLA-A2-restricted CD8$^+$ $\alpha\beta$ T cell clone, which produces IFN-γ in response to M1p58–66-presenting, HLA-A2$^+$ APCs (labeling and gating strategy of the IFN-γ assay is

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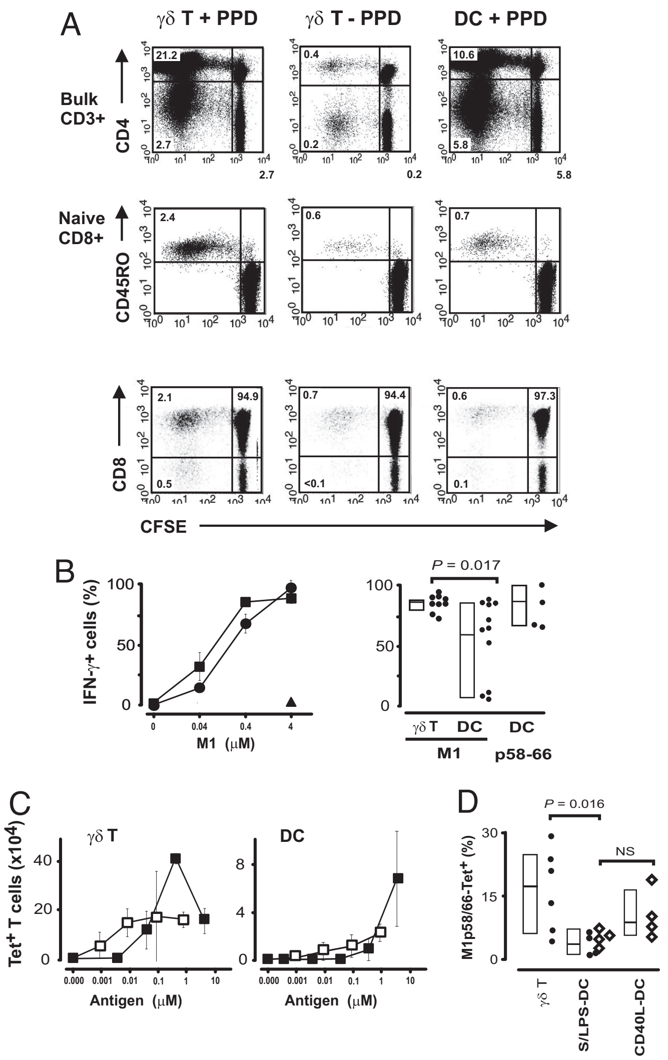

Fig. 1. γδ T-APCs cross-present soluble protein antigen to CD8$^+$ $\alpha\beta$ T cells. (*A*) γδ T-APCs and DCs were treated with PPD, then washed and cocultured for 10 days with CFSE-labeled bulk $\alpha\beta$ T cells or purified naïve CD8$^+$ $\alpha\beta$ T cells at a APC/responder cell ratio of 1:10. Results in row 3 illustrate that the majority of proliferating CD45RO$^+$ cells were CD8$^+$ responder cells. Data are representative of 2 and 3 experiments with bulk and naive CD8$^+$ $\alpha\beta$ T cells, respectively. (*B*) γδ T-APCs (squares) and DCs (circles) cross-present influenza matrix protein M1 to the HLA-A2-restricted, M1p58–66-specific CD8$^+$ $\alpha\beta$ T cell clone FLUMA55 (APC/responder cell ratios varied between 1: 5 and 3:1 but this variation had no obvious effect on the results with FLUMA55). Negative control, 4 μM M1 treated, HLA-A2-negative B cells (triangles). The *Right* compiles data from 7 independent FLUMA55 cross-priming experiments with γδ T-APCs and DCs treated with 0.4 μM M1; additional control, 0.1 μM M1p58–66 pulsed DCs. Boxes' lower/upper ends and middle lines depict 25/75 percentile and median. (*C*) Bulk CD8$^+$ $\alpha\beta$ T cells were stimulated with M1 (filled squares) or M1p58–66 (open squares) treated γδ T-APCs and S/LPS-DCs (APC/responder cell ratio of 1:20), and, after 10 days of culture, M1p58–66-specific responder cells were quantified by M1p58–66-tetramer staining. (*D*) γδ T-APCs and DCs, either treated with shear force and LPS or with CD40L, differ in their efficiency to cross-present M1 to bulk CD8$^+$ $\alpha\beta$ T cells. Blood cells from 2 to 4 different donors; 1-tailed students *t* test; NS, not significant.

explained in Fig. S1). M1 pretreated γδ T-APCs induced robust and highly reproducible effector cell activation, and responses were already detected when 0.04 μM M1 were used during APC preparation (Fig. 1*B*). These findings did not result from a potential M1p58–66 peptide contamination in the M1 protein preparation (Fig. S2), demonstrating that γδ T-APCs were able to take up and process exogenous M1 for presentation in the context of MHC I molecules. HLA-mismatched B cells used as feeder cells during in vitro activation of Vγ9Vδ2$^+$ T cells failed to cross-present M1 (Fig. 1*B*). Of note, γδ T-APCs from different donors gave reproducible results, which is in contrast to the strikingly variable responses obtained with DCs (Fig. 1*B*). Of interest, γδ T-APCs were able to take up and process M1 protein over a wide range of culture time and still showed antigen presentation function after prolonged culture in the absence of antigen (Figs. S3 and S4).

In the next step, we tested M1p58–66-pulsed γδ T-APCs for their ability to induce proliferation in blood CD8$^+$ $\alpha\beta$ T cells. M1p58–66-specific cells (0.01–0.5%), assessed by M1p58–66-tetramer staining, are primarily found in the memory T cell compartment of healthy HLA-A2$^+$ individuals (14). Responses obtained with M1p58–66-pulsed γδ T-APCs were unmatched in terms of potency and efficacy, as compared with DCs, monocytes and B cells (Fig. S5). Moreover, γδ T-APCs were also very adept in cross-presentation of M1, involving the uptake and intracellular processing of exogenous protein, to this polyclonal M1p58–66-reactive CD8$^+$ $\alpha\beta$ T cell compartment (Fig. 1*C*). Striking variation in responses to DCs prompted us to evaluate different strategies for DC generation, including substituting IL-15 for IL-4 during monocyte differentiation (data not shown), and applying CD40-signaling as opposed to shear force in combination with LPS to induce DC maturation. None of these treatments led to substantial improvements (Fig. 1*D*), and in all subsequent experiments shear force/LPS-treated DCs were used.

Antigen Cross-Presentation by γδ T-APCs Involves Proteasome Activity and de Novo Synthesized MHC I Molecules. The route(s) of antigen processing leading to peptide loading onto MHC I within γδ T-APCs are not known. The 2 inhibitors lactacystin and brefeldin A selectively target the proteasome and the transGolgi network, respectively, and thus interfere with the classical (proteasome- and protein export-dependent) MHC I pathway. We found that cross-presentation in γδ T-APCs and DCs was fully inhibited by these compounds, supporting the notion that γδ T-APCs do not differ from DCs in their use of the classical MHC I pathway for processing of exogenous influenza matrix protein M1 (Fig. S6).

Because de novo MHC I synthesis is of primary importance for induction of CD8$^+$ $\alpha\beta$ T cell responses (15), we performed immunocytochemical analysis of resting and activated Vγ9Vδ2$^+$ T cells. TCR-triggered up-regulation of MHC I was substantial, paralleled blast formation and was composed of increased intracellular and cell surface MHC I staining (Fig. 2*A*). Peak levels in total MHC I staining were >7-fold above levels in unstimulated γδ T cells and were reached between 18 h and 48 h of culture. As expected (16), shear force and LPS treatment in DCs resulted also in increased cell surface MHC I expression. These findings were confirmed by flow cytometric analysis of MHC I in resting versus activated γδ T cells (Fig. 2*B*). Elevated cell surface staining was due to de novo MHC I synthesis as evidenced by lack of intracellular MHC I storage compartments in resting γδ T cells and by sustained colocalization of MHC I with the transGolgi network (GM130) during the course of stimulation (Fig. 2*C* and Fig. S7).

Immunoproteasome in γδ T-APCs Prevents Induction of Melp26–35-Specific CD8$^+$ $\alpha\beta$ T Cell Responses. To test a potential function in anti-tumor immunity, we next studied the ability of γδ T-APCs to cross-present the melanocyte/melanoma-differentiation antigen Melan-A (MART-1), which contains the immunodominant peptide Melp26–35 recognized by HLA-A2-restricted CD8$^+$ $\alpha\beta$ T cells (17). Melp26–35 specific CD8$^+$ $\alpha\beta$ T cells are readily detected in both melanoma patients and healthy individuals (14), thus allowing us to study Melan-A cross-presentation by γδ T-APCs with blood cells from healthy volunteers. Of note, Melan-A-pretreated γδ T-APCs and DCs both failed to induce IFN-γ production in HLA-A2-restricted, Melp26–35-specific

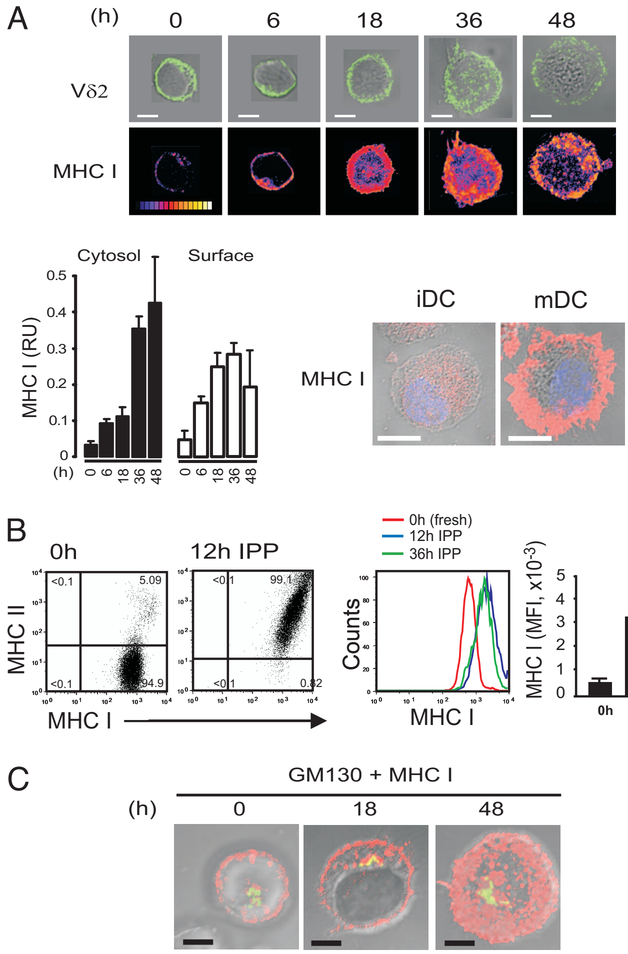

Fig. 2. Cellular distribution of MHC I during activation of $V\gamma9V\delta2^{+}$ T cells. (*A*) Activation of $V\gamma9V\delta2^{+}$ T cells with IPP for 6–48 h in the presence of feeder B cells followed by confocal immunofluorescence microscopic analysis of Vδ2-TCR staining (green) in combination with digital interference contrast images (*Upper*), or with MHC I staining (fire scale color mapping) (*Lower*); 0 h, resting γδ T cells. Control, digital interference contrast images in combination with MHC I (red) and nuclei (blue) stainings in immature (iDC) and mature DCs (mDC). Bar graph represents quantifications of intracellular (cytosol) and cell membrane (surface) associated MHC I within individual γδ T cells at the indicated IPP stimulation time points; relative unit (RU) of 1 equals 10^{6} counts with 3–6 cells analyzed per data point. (*B*) Cell surface expression of MHC I and MHC II was analyzed by flow cytometry in freshly isolated (nonstimulated) and activated $V\delta2^{+}$ γδ T cells that were stimulated for 12 or 36 h with IPP. (*C*) Increased cell surface MHC I staining in γδ T cells involves de novo MHC I synthesis. MHC I (red) in conjunction with GM130 (green) is shown as maximum intensity projections in combination with digital interference contrast images (50:50 fluorescence intensity ratio in yellow). [Scale bars: 5 μm (10 μm for DCs).]

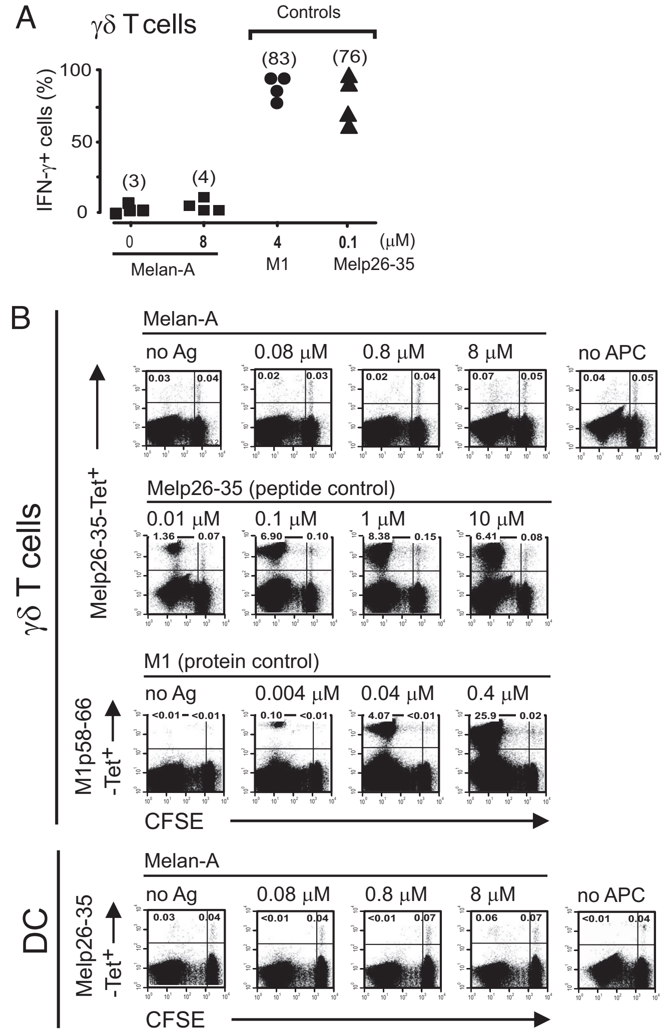

Fig. 3. γδ T-APCs and DCs fail to cross-present Melan-A to Melp26–35-specific $CD8^{+}$ αβ T cells. (*A*) γδ T-APCs were treated with or without Melan-A and then cocultured with the HLA-A2-restricted, Melp26–35-specific responder cell clone LAU 337 for determination of intracellular IFN-γ. Controls include Melp26–35-pulsed γδ T-APCs together with LAU 337 responder cells, and M1 cross-presenting γδ T-APCs together with FLUMA55 responder cells. Numbers in brackets represent the mean. (*B*) γδ T-APCs and DCs were incubated with Melan-A at indicated concentrations and cocultured with CFSE-labeled, HLA-A2-restricted blood $CD8^{+}$ αβ T cells at a APC/responder cell ratio of 1:10. Alternatively, Melp26–35 pulsed γδ T-APCs or M1 cross-presenting γδ T-APCs were used and the numbers (percentage of total) of Melp26–35- and M1p58–66-tetramer positive responder cells were determined at 10 days of culture. Data are representative of 2–4 experiments.

responder cell clones (Fig. 3*A*, and data not shown). This failure was not due to problems with antigen presentation per se or due to a weak responsiveness by the responder clone because Melp26–35-pulsed γδ T-APCs and DCs induced strong IFN-γ responses. Moreover, uptake of soluble proteins was not affected either because the same γδ T-APC preparations were perfectly capable of cross-presenting M1 to the M1p58–66-specific responder cell clone (Figs. 1*B* and 3*A*). These findings were mirrored in a responder cell proliferation assay, showing that Melan-A pretreated γδ T-APCs or DCs failed to induce the expansion of Melp26–35-tetramer^{+} cells present within bulk $CD8^{+}$ αβ T cells (Fig. 3*B*). Again, control APCs, including Melp26–35-pulsed γδ T-APCs and M1 cross-presenting γδ T-APCs, performed well. These findings illustrate that lack of Melan-A cross-presentation was neither due to problems with antigen uptake or processing per se nor peptide presentation and recognition by peptide-specific $CD8^{+}$ responder cells.

The proteasome exerts a crucial role in the classical MHC I pathway of peptide presentation and exists in 2 forms, the standard proteasome present in all nucleated cells and the immunoproteasome, which contains alternative, IFN-γ- or TNF-α-inducible protease subunits (18). The immunoproteasome produces a different spectrum of peptides and thereby influences the shape of $CD8^{+}$ αβ T cell responses under inflammatory conditions. For instance, it has been shown that the immunodominant peptide Melp26–35 is readily produced by the standard proteasome whereas it is rapidly degraded by the immu-

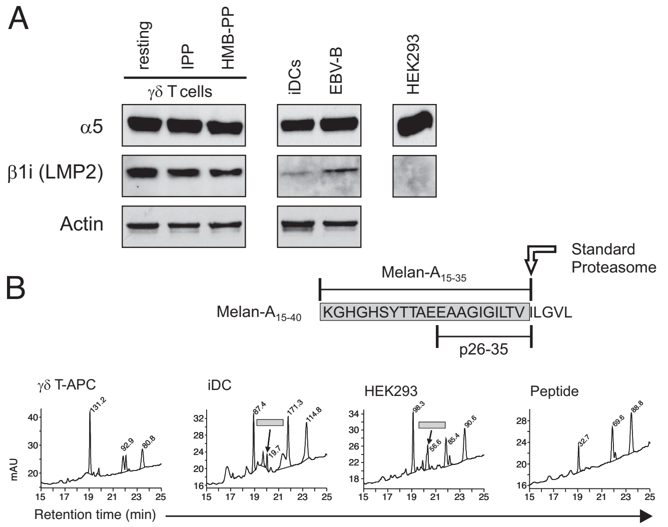

Fig. 4. Vγ9Vδ2$^+$ T cells express highly active immunoproteasome. (*A*) Proteins in lysates of freshly isolated (resting) Vγ9Vδ2$^+$ T cells or γδ T-APCs or monocyte-derived DCs (iDCs) or B cells (EBV-B) were separated by SDS/PAGE and analyzed by Western blot. α5, protease subunit present in both standard and immunoproteasome; β1i (LMP2), immunoproteasome-specific subunit; Actin, protein loading control. (*B*) Purified proteasome from γδ T-APCs, monocyte-derived immature DCs (iDCs) and human embryonic epithelial cells (HEK293) were incubated at 37 °C for 16 h with the peptide substrate Melan-A$_{15-40}$ and the peptide products were fractionated by reverse-phase HPLC and then identified by mass spectroscopy. The peaks at 20.3 min elution time contained the standard proteasome-specific peptide Melan-A$_{15-35}$ (highlighted with gray bars). The yield in Melan-A$_{15-35}$ was highest with proteasome preparations from HEK293 cells. Of note, the Melan-A$_{15-35}$ was not detected with proteasome preparations from γδ T-APCs, suggesting dominant proteolytic activity by the immunoproteasome. Data are representative of 2 and 3 separate experiments.

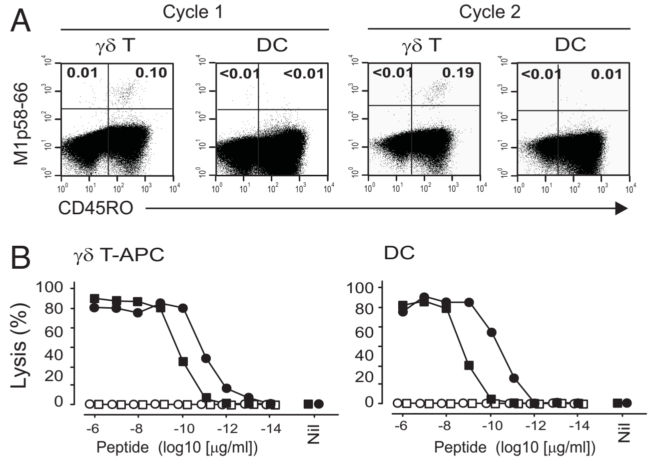

Fig. 5. Cross-presenting γδ T-APCs induce robust primary CD8$^+$ αβ T cell responses. (*A*) γδ T cells and DCs, treated with 4 μM M1 (see Fig. 1*C*), were cultured with sorted naïve CD8$^+$ αβ T cells (APC/responder cell ratio of 1:20) for 10 days (cycle 1), or were restimulated with M1 cross-presenting APCs and cultured for another 10 days (cycle 2). Responder cells were identified by M1p58–66-tetramer staining. (*B*) M1 cross-presenting γδ T-APCs and DCs (*Left* and *Right*, respectively) were used as APCs, and naïve CD8$^+$ αβ T cell-derived responder cells were cloned by limited dilution culture. ^{51}Cr-labeled target cells were pulsed with M1p58–66 (filled circles and squares) or unrelated Melp26–35 (open circles and squares) at indicated concentrations or were unpulsed and mixed at a 1:1 ratio with responder clones. One representative high-affinity (circles) and low-affinity (squares) responder clone are shown for each cytotoxic T cell cloning experiment; data are representative of 26 M1p58–66-tetramer$^+$ T cell clones.

noproteasome (19, 20). We found that peripheral blood γδ T cells and in vitro generated γδ T-APCs contained predominantly the immunoproteasome (Fig. 4*A*). Western blot analysis revealed the relative amount of immunoproteasome (specific subunit β1i/LMP2) in relation to the total amount of proteasome (common subunit α5) (21). Immature DCs and B cells had much lower amounts of the immunoproteasome, and HEK293 cells served as a standard proteasome control. The immunoproteasome in γδ T-APCs was functionally predominant as demonstrated by peptide product analysis after digestion of the peptide substrate Melan-A$_{15-40}$ with freshly prepared, purified proteasome (Fig. 4*B*). This was not the case for immature DCs, where the standard proteasome-resistant (but immunoproteasome-sensitive) signature peptide fragment Melan-A$_{15-35}$ was readily observed. As expected, immunoproteasome-negative HEK293 cells also produced the Melan-A$_{15-35}$ peptide. Collectively, the predominant immunoproteasome activity in γδ T-APCs fully agrees with the complete absence of Melp26–35-specific CD8$^+$ αβ T cell responses in our Melan-A cross-presentation assays (Fig. 3).

γδ T-APCs Induce Effector Cell Differentiation in Naïve CD8$^+$ αβ T Cells. To examine whether γδ T-APCs have professional cross-presentation capabilities, M1 cross-presenting γδ T-APCs or DCs were cultured with a 20-fold excess of sorted autologous naïve CD8$^+$ αβ T cells (>98% purity; Fig. S8). M1p58–66-specific responder cells were quantified after 10 days of culture (cycle 1) or after a second round of stimulation (cycle 2). After cycle 1 a significant portion of CD8$^+$ αβ T cells expressed the memory marker CD45RO (Fig. 5*A*). M1p58–66-specific T cells became detectable (0.1–0.3% among total CD8$^+$ αβ T cells), as assessed by tetramer staining, and this T cell subset was maintained during secondary expansion, permitting their further examination (see below). Proliferation responses were remarkable, because the frequency of M1p58–66-specific (M1p58–66-tetramer$^+$) cells in the starting population of naïve blood CD8$^+$ T cells was below the level of detection (<1/50,000) (22). In contrast to γδ T-APCs, the responses of naïve CD8$^+$ αβ T cells to M1 cross-presenting DCs were highly variable or undetectable (example in Fig. 5*A*). Specificity of the M1p58–66 response is evidenced by the lack of tetramer staining in (*i*) cultures without APCs and (*ii*) cultures with γδ T-APCs and DCs cross-presenting the irrelevant antigen Melan-A (data not shown).

After the second cycle of stimulation with M1 cross-presenting APCs, 21% of sorted M1p58–66-tetramer$^+$ T cells carried Vβ17-TCRs, indicating that most of the sorted responder cells derived from naïve precursors (22, 23). As expected, the fraction of Vβ17$^+$ cells increased to >70% during bulk culture (data not shown). For further analysis, M1p58–66-tetramer$^+$ sorted cells were cloned by limited dilution. Twenty-six T cell clones were M1p58–66-tetramer$^+$, and all of these specifically lysed M1p58–66 pulsed target cells with half maximal effective M1p58–66 concentrations ranging between 10^{-9} and 10^{-11} μg of peptide/mL (Fig. 5*B*). In support of specificity, target cells either unpulsed or pulsed with the unrelated Melan-A peptide Melp26–35 were not recognized (Fig. 5*B*). Similar results were obtained with DCs in experiments where numbers of induced M1p58–66-tetramer$^+$ CD8$^+$ αβ T cells were large enough for cloning and further analysis (Fig. 5*B*). Collectively, these data demonstrate that cross-presenting γδ T-APCs were capable of triggering naïve CD8$^+$ αβ T cell proliferation and effector cell generation.

Discussion

Many exceptional properties distinguish Vγ9Vδ2$^+$ T cells from αβ T cells. For instance, Vγ9Vδ2$^+$ T cells lack the coreceptors CD8 and CD4, which restrict antigen recognition in αβ T cells to peptides that are presented in conjunction with MHC I and

MHC II molecules, respectively. As a consequence, this allows selectivity for nonpeptide antigens without impairing signal strength (24) and, at the same time, releases the constraint for the need of conventional APCs for induction of Vγ9Vδ2$^+$ T cell responses. A second distinguishing feature of Vγ9Vδ2$^+$ T cells is their broad, polyclonal activation by a single class of nonpeptide ligands derived from microbes or stressed tissue cells with alternative or aberrant isoprenoid metabolism (7, 8). This enables the immediate engagement of a large number of Vγ9Vδ2$^+$ T cells (up to 10% of total blood T cells) in response to infections or tumors where such nonpeptide ligands are produced. By comparison, the frequency of circulating $\alpha\beta$ T cells with selectivity for a single peptide-MHC complex is $>10^4$-fold lower. Responses of Vγ9Vδ2$^+$ T cells to HMB-PP and related compounds are both rapid and vigorous (25) and, thus, are reminiscent of cellular responses mediated by receptors for pathogen-associated molecular patterns. It is not known how Vγ9Vδ2$^+$ T cells recognize these nonpeptide agonists and whether TCR triggering requires the presentation of such compounds by specialized "feeder cells." Importantly, transition from resting to fully activated Vγ9Vδ2$^+$ T cells (termed $\gamma\delta$ T-APCs) is associated with the expression of CCR7 that enables lymph node homing and a plethora of antigen-presentation and costimulation molecules (10, 26). It is uncertain where in the human body antigen presentation by $\gamma\delta$ T-APCs may take place, but possible sites include the site of microbial encounter in peripheral tissues and infection draining lymphoid tissues (9, 10, 25, 27–29).

We demonstrate here that $\gamma\delta$ T-APCs were capable of processing exogenous soluble proteins and presenting peptide-MHC I complexes to antigen-specific CD8$^+$ $\alpha\beta$ T cells. $\gamma\delta$ T-APCs also triggered naïve CD8$^+$ $\alpha\beta$ T cell proliferation and effector cell generation, a process known to depend on professional APCs (1). Surprisingly, $\gamma\delta$ T-APCs were much more reliable than monocyte-derived DCs in terms of effectiveness and reproducibility. Changes in the preparation of monocyte-derived DCs, for instance by substituting IL-15 for IL-4 during monocyte differentiation or by including alternative DC maturation stimuli, did not improve their performance. These difficulties were not observed in the induction of CD4$^+$ $\alpha\beta$ T cells responses (10), pointing toward some critical factors in the in vitro preparation of monocyte-derived DCs that specifically affect antigen cross-presentation. Maturation dependent and independent processes have been shown to downmodulate antigen cross-presentation in DCs (30, 31). We consider the reliable performance in antigen cross-presentation an important feature of $\gamma\delta$ T-APCs.

Our current knowledge supports a model whereby $\gamma\delta$ T-APCs are induced from peripheral blood Vγ9Vδ2$^+$ T cells after their recruitment to the site of infection in response to local inflammatory chemokines (9, 32) and in response to their exposure to microbe-derived agonists, such as HMB-PP (7, 10, 33). Positioning in peripheral blood and immediate responsiveness to inflammatory cues ensure their rapid, innate-like involvement in host defense. $\gamma\delta$ T-APCs not only mobilize proinflammatory (IFN-γ, TNF-α, chemokines) and cytotoxic activities (4–6) but also process microbial antigen for induction of CD8$^+$ (as evidenced here) and CD4$^+$ $\alpha\beta$ T cell responses (10–13). This model portrays $\gamma\delta$ T cells as forming a vital part in the first-line defense in response to microbial challenges or tumors and emphasizes their exceptional ability to bridge innate and adaptive immunity. Collectively, the extraordinary ability to process extracellular antigen for induction of cytotoxic T cells provides the framework for studying the usefulness of human $\gamma\delta$ T-APCs in immunotherapy.

Materials and Methods

Cell Isolation and APC Preparation. Human PBMCs of HLA-A2-positive (subtype *0201) donors were used to isolate $\gamma\delta$ T cells, CD14^{high} cells and CD8$^+$ $\alpha\beta$ T cells by positive or negative selection, respectively, and B cells by negative selection, using the magnetic cell sorting system from Miltenyi Biotec (10). Positive selected $\gamma\delta$ T cells were stimulated with 50 μM isopentenyl pyrophosphate (IPP) preparation (see *SI Material and Methods*) (Sigma–Aldrich) presented by either autologous primary B cells or HLA-A2-negative EBV-B cell lines (irradiated with 40Gy or 100–120Gy, respectively, followed by washing) in round-bottom 96-well plates (0.2–1 $\times$ 10^6 $\gamma\delta$ T cells per well) for 18 h in medium supplemented with human serum plus 10 ng/mL IL-15. Alterations in DCs preparation included culturing of CD14^{high} cells in 100 ng/mL IL-15 and 50 ng/mL GM-CSF as opposed to the standard procedure involving 10 ng/mL IL-4 and 50 ng/mL GM-CSF for 6–7 days. Maturation for 8 h was initiated by applying shear force (cluster disruption by pipetting) and 1 μg/mL LPS (from Salmonella abortus equi, Sigma) or by culturing of DCs with CD40L-expressing J558L cells at a 2:5 ratio (CD40L DC). APCs were washed 3 times before use in functional assays. Additional APCs included fresh CD14^{high} monocytes and autologous, γ-irradiated (40 Gy) feeder B cells treated with IPP for 18 h before use ("IPP-BC" control).

Antigens and inhibitors (Brefeldin A and Lactacystin) were added at indicated concentrations 2 h and 4 h, respectively, before start of $\gamma\delta$ T-APC preparation or induction of DC maturation. For peptide pulsing, APCs including $\gamma\delta$ T-APCs, 5 h maturated DCs, monocytes or control B cells were incubated for 3 h with the peptides in the serum-free medium. All APCs were irradiated ($\gamma\delta$ T-APCs 9–10Gy; monocytes 26Gy; DCs 30Gy; IPP-BC control 9–10Gy) and then washed 3 times before use.

Negatively magnetic beads sorted naïve CD8$^+$ $\alpha\beta$ T cells excuded cells expressing VγVδ-TCR, CD1c, CD4, CD11b, CD11c, CD14, CD16, CD19, CD25, CD45RO, CD56, CD64, HLA-DR, CD138, CCR5 and CXCR3, whereas bulk CD8$^+$ $\alpha\beta$ T cells were negative for VγVδ-TCR, CD1c, CD4, CD14, CD16, CD19, CD25, CD64, HLA-DR, and CD138 (10). Carboxyfluorescein Diacetate Succinimidyl Ester (CFSE) labeling was performed as described in ref. 10.

Antigen Presentation Assays. Assay for PPD-dependent CD8$^+$ T cell proliferation included CFSE-labeled bulk $\alpha\beta$ T cells or purified naïve CD8$^+$ T cells together with APCs, and the cells were cultured in the absence of IL-2. Cross-presentation of Influenza Matrix M1 and Melan-A protein was done on a HLA-A*0201 background, and responder cells were identified after 10 days of coculture in the presence of IL-2 (20 units/mL to bulk and 200 units/mL to naïve CD8$^+$ cultures) by tetramer staining (14). IL-2 was only added 48 h after coculture onset. In addition, M1p58–66 tetramer binding cells derived from naïve CD8$^+$ $\alpha\beta$ T cell preparations were sorted, further cultured or cloned by limiting-dilution for further analysis. Additional assays involved intracellular detection of IFNγ in CD8$^+$ $\alpha\beta$ T cell clones specific for the relevant peptides (FLUMA55 for M1p58–66, and LAU337 6B7 for Melp26–35). IFNγ was detected in responder cells by flow cytometry after coculture with antigen-pretreated APCs. BrefeldinA was added 30–45 min after initiation of cocultures, and 5–6 h later, cells were washed twice in FACS-buffer and subjected to a FC-Block (excess of human IgG in FACS-buffer) for 15 min. Cells were then stained for intracellular IFNγ as described in Fig. S1.

Proteasome Studies. For the detection of proteasome subunits, cell lysates from $\gamma\delta$ T-APCs, either freshly isolated from blood or stimulated for 24 h with IPP or HMB-PP, or immature, monocyte-derived DCs or an EBV-B cell line or human embryonic kidney HEK293 cells were separated by SDS-12% PAGE and subjected to Western blot analysis (19). The subunit α5, a common subunit of both the standard and immunoproteasome, and β1i (LMP2), a immunoproteasome-specific subunit, were detected with specific antibodies. Staining of HEK293 extract proteins was included as negative control for the immunoproteasome. For functional studies, proteasomes were immunopurified from extracts from $\gamma\delta$ T-APCs, immature DCs and HEK293 cells as described in ref. 21. Proteasomes were eluted and directly incubated at 37 °C for 16 h with 4 μg of synthetic peptide Melan-A$_{15-40}$. As control, peptide Melan-A$_{15-40}$ was incubated under the same conditions in the absence of proteasomes. The material was separated by reverse phase HPLC and the peptides within the peak fractions were identified by mass spectrometry as described in ref. 19. The C terminus of the antigenic peptide Melan-A$_{26-35}$ is produced by the standard proteasome upon cleavage of the Melan-A$_{15-40}$ peptide substrate, and absence of this peptide intermediate indicates standard proteasome-independent processing.

Confocal Microscopy. Immunostaining of paraformaldehyde-fixed cytospins of PBMCs, $\gamma\delta$ T cells and monocyte-derived DCs was carried out essentially as described in ref. 10. In brief, 1% saponin permeabilized cytospins were blocked with 3 mg/mL human Ig and casein sodium salt, and then stained with labeled anti-human HLA-ABC-Alexafluor647 (clone w6/32, mIgG2a, BioLegend) and primary antibodies against Vδ2-TCR (clone BB3, mIgG1; gift from M. B. Brenner) followed by treatment with fluorescently labeled goat anti-mouse IgG1-Alexafluor488 (Molecular Probes), and finally mounted in Prolong Gold (Molecular Probes). For triple stainings, FITC-labeled anti-human

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GM130 (clone35, BD Transduction Laboratories) was applied together with directly labeled anti-human Vδ2-TCR (clone B6.1, BD PharMingen) and anti-human HLA-ABC-Alexafluor647. Stacks of confocal images (scaling resolution: 0.06 μm $\times$ 0.06 μm $\times$ 0.15 μm) of the samples were acquired with the laser-scanning microscope LSM 510Meta (Zeiss), processed by Huygens essential deconvolution software (Scientifique Volume Imaging) and analyzed using 3D-image restoration software package Imaris 5.5 (Bitplane). For subcellular MHC I (HLA-ABC) localization and quantification in IPP-activated Vγ9Vδ2$^+$ T cells, fluorescence intensities of defined spheres (0.3 μm diameter; threshold 100 counts) within cell surface membrane, cytoplasma and nuclei (negative control) were measured in 3D-restored images. Fluorescence intensities (relative unit [RU] of 1 equals 10^6 counts) associated with the respective cell compartments were determined per cell by the spot function.

Media, reagents, and antibodies are described in *SI Material and Methods*.

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